



PTH 6120: NEUROSCIENCE

Fall 2026

T, Th 10:30 - 1:00

SRH 114 and 215

Department of Physical Therapy

Instructor:

Dr. Barbara Crane

Email: bacrane1@plymouth.edu

Phone: (603) 535-2897 (on campus 5-2897)

Office: 229 Samuel Read Hall

Office hours: TR 3:30-4:30; F 12-2

or by appointment

Teaching Assistants:

Trish Feeney

mpf1038@usnh.edu

(770) 362-7811

Connor Belt

cjb1148@plymouth.edu

(443) 716 - 6666

Course Information

Semester: Fall

Credit Hours: 4

Weekly Clock Hours: 5

Semester Clock Hours: 70

Course schedule: Tuesday, Thursday 10:30 – 1:00

Location: Sam Read Hall room 114 and 215 (for lab activities)

Pre-Requisites

Clinical & Functional Anatomy, Clinical Physiology, Patient/ Client Management I, Ethics & Value Systems, Integrated Clinical I

Co-Requisites

Dynamic Systems I, Clinical Inquiry I: Causality & Inference, Musculoskeletal Conditions & Management, Patient / Client Management II, Integrated Clinical II

Course Description

Covers structure and function of the nervous system. Relates anatomy and physiology to functional capabilities related to movement and cognition; considers the impact of lesions, and the causes of observed signs and symptoms during development, adaptations and trauma through the life span.

Students apply neuroscience to reasoning about clinical scenarios and to the analysis of patient/client problems.

Student Learning Outcomes

1. Describe the anatomical components and relationships of the nervous system.
2. Demonstrate understanding of the cells that make up the structure and contribute to the functions of the nervous system; including the roles of neurons, glia, and neural stem cells in the developing and adult central and peripheral nervous system.
3. Describe the development of the nervous system.
4. Discuss how development impacts function of the nervous system across the lifespan.
5. Describe and discuss the physiology of the nervous system with a primary emphasis on sensory, movement and cognitive systems.
6. Demonstrate the ability to apply the basic concepts of physiology to the analysis of patient/client problems related to the neurological system.
7. Select and administer age-appropriate tests and measures to assess the function of the neurological system; including tests that allow one to measure: cranial and peripheral nerve integrity, reflexes, sensory integrity, pain, balance, and basic mental functions (e.g. arousal, attention and cognition).
8. Relate the International Classification of Function (ICF) to describe how neurological systems emerge as function and consider the whole system impact of dysfunction (impairments).
9. Interpret common diagnostic images used in differential diagnosis of neurological conditions including CT, MRI, fMRI, PET to better understand the problems experienced by patients with conditions affecting the nervous system.

Textbooks:

Reading assignments will be provided from three Neuroscience textbooks in the *Access Physiotherapy* resources:

Amthor FR, Theibert AB, Standaert DG, Roberson ED. eds. *Essentials of Modern Neuroscience*. McGraw Hill; 2020.

<https://accessphysiotherapy.mhmedical.com/book.aspx?bookid=2938>

Waxman SG. ed. *Clinical Neuroanatomy, 30th Edition*. McGraw Hill; 2024.

<https://accessphysiotherapy.mhmedical.com/book.aspx?bookid=3408>

Mosconi T, Graham V. eds. *Neuroscience for Rehabilitation*. McGraw-Hill Education; 2017.

<https://accessphysiotherapy.mhmedical.com/book.aspx?bookid=2258>

Additional readings and materials will also be assigned (see Canvas for details).

Other suggested learning resources:

1. There are several resources indicated in the “Neuroscience Course Resources” module in Canvas.
2. Physio U “Neuroanatomy” App: <https://app.clinicalpattern.com/neuroanatomy/>
3. Learn Medical Neuroscience (online learning resource): <http://www.learnmedicalneuroscience.nl/>
4. “Human Brain Coloring Book” (available on Amazon as well as other sites)
5. Picmonic videos (as assigned in Canvas)

iPad Access

There are 15 iPads stored in the closet in the Anatomage room. You are free to use them in SRH (do not remove from the building) to assist your learning of course material. You can use them to access your Physio U account and the following apps are also available and may be of use:

- | | |
|-----------------------------|-------------------|
| 1. 3D Brain | 4. Brain Tutor 3D |
| 2. The Complete Human Brain | 5. Nerve Whiz |
| 3. Neurolocator | 6. Brain Anatomy |

Description of teaching methods and learning experiences

Readings, discussions, tutorials, videos, laboratory activities, case discussions, clinical problems

Assignments and Activities

Weekly quizzes*	55%
Comprehensive Final Exam	35%
Neuro exam skills, comprehension and interpretation (OSCE/Practical)	10%

* Students will be permitted to drop the lowest scoring quiz from their total quiz grade (1 only)

Student Evaluation/Grading

A	94.0 – 100
A-	90.0 – 93.9
B+	87.0 – 89.9
B	83.0 – 86.9
B-	80.0 – 82.9
C+	77.0 – 79.9
C	73.0 – 76.9
F	below 73.0

Students are reminded that continuation in the program necessitates achieving an overall grade point average of 3.0 (B) or better. Grades below 2.0 (C) are not acceptable or counted toward the DPT degree. No more than 9 credits below 3.0 may be counted toward the DPT degree (grades of B-, C+, or C). Make-up examinations will be given only in specific cases of absence (documented absences). Circumstances that may warrant an absence should be discussed with the instructor before the scheduled time of the exam. Failure to notify the instructor as stated may result in a zero for the exam.

Professional Conduct: Students are reminded of the program expectation that they follow Professional Conduct; see the Student Handbook Professional Conduct and Reasonable Professional Expectations for the process of handling violations, remediation and consequences.

Cell Phones: Cell phone use (including texting) is prohibited in class. Also, please respect your classmates by turning your phone to vibrate prior to class start time. Do not turn phones off as the University uses texts to relate emergency information.

Graduate Academic Policies: The DPT Program is a selective admissions cohort-structured program.

Some Graduate Academic Policies may not be relevant to the program. Students should refer to the Doctor of Physical Therapy Program Handbook for specific information about Scheduling, Capstones, and Internships. The sections on Grading System and Academic Standing Policy do apply to DPT students: <https://coursecatalog.plymouth.edu/university-policies-procedures/graduate-academic-policies/>

Fair Grading Policy: Fair and equitable grading reflects values to which all members of the Plymouth State community commit themselves. Grades are used to assess the relative extent to which students achieve course objectives in all for-credit courses at PSU. Academic freedom allows instructors to determine course objectives within the bounds of established curricula, and the means by which a student's mastery of those objectives will be evaluated; and evaluate the quality of work on individual exams or assignments. Information can be found at <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Fair Grading.

Student Support: Student Support Foundation (SSF) provides short-term emergency financial assistance and long-term student support. For more information, see: <https://campus.plymouth.edu/student-support-foundation/> Student Support Foundation also runs a food pantry, located in Belknap Hall. To learn more about SSF or access the food pantry, either via open hours or a private appointment, contact the SSF advisor, at psu-ssf@plymouth.edu.

Academic Dishonesty: Cheating and plagiarism are very serious forms of academic dishonesty. Any use of unauthorized assistance (including assistance from AI applications) on exams, papers, homework assignments, or other course work constitutes cheating. Knowingly providing assistance during an exam or allowing other students to copy one's work is also a serious form of academic dishonesty. Plagiarism consists of submitting written work that has been developed wholly or partially by someone else. Submitting written work in which the ideas of others have been duplicated or even paraphrased without proper reference to the author is also a form of plagiarism. Also considered plagiarism is the acquisition of term papers or other assignments from another source and subsequent presentation of these materials as the student's own work. In addition, students may not use papers in more than one course without the permission of both instructors.

Any student suspected of academic dishonesty will be reported to the Program Director immediately. A full description of academic integrity, violations and procedures when a violation to academic integrity in a course is suspected can be found at: <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Academic Integrity.

Attendance Policy: Attendance is expected to all class sessions. Students are ultimately responsible for their attendance, however faculty will assess students Professional Expectations based, in part, on their attendance (See Student Evaluation/Grading). Students who miss class for any reason are responsible for the information and work missed. Students should speak with the professor before missing the scheduled class when able.

Campus Accessibility Policy: Plymouth State University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you should immediately contact Campus Accessibility Services (CAS) in Speare210 (535-3300) to determine whether you are eligible for such accommodations. Academic accommodations will only be considered for students who have registered with CAS. If you

have a Letter of Accommodation for this course from CAS, please provide the instructor with that information privately so that you and the instructor can review those accommodations:

<https://campus.plymouth.edu/accessibility-services/>

Diversity and Inclusivity Statement:

We consider this classroom to be a place where you will be treated with respect, and we welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. See this web page for the Plymouth State University IDEA Center Inclusion, Diversity, Equity, and Access. <https://www.plymouth.edu/idea-center>

Tentative Class Schedule (this schedule is subject to change throughout the semester and students will be notified of any changes made).

Details including specific reading requirements, assignments and activities will be provided on the course Canvas page.

Week	Topics
1	Course introduction; Introduction to neuroscience/ overview of the nervous system; Cerebrum: surface anatomy and functional areas; Cells and synapses and their roles in the PNS and CNS Lab: Neuro Examination – Overview and taking a patient history, role of history and examination in differential diagnosis; systems review as part of neuro exam
2	Development of the nervous system; Internal Anatomy of the forebrain (telencephalon and diencephalon); Radiology/diagnostic imaging of the NS – CT, MRI, fMRI, PET, Angiography Lab: Intro to case format; mental status examination – (alertness, attention, language, etc.)
3	Surface Anatomy of the Brainstem; Cranial nerves; Internal Anatomy of the Brainstem – midbrain and pons Lab: Cranial nerve testing/differential diagnosis and clinical applications
4	Internal anatomy of brainstem – medulla; Spinal Cord – overview; Pathways – spinal reflex arc Lab: Assessment of spinal reflexes
5	Spinal cord – internal anatomy; Sensory receptors; begin discussion of ascending and descending pathways Lab: Assessment of the somatosensory system and clinical applications (light touch, vibration, sharp/dull, temperature)
6	Somatosensory system – receptors and pathways (cont.); Pain neuroscience Lab: Clinical assessment of pain; assessing the sensory pathways to the CNS; Sensory processing – graphesthesia, stereognosis, sensory extinction
7	Descending motor system (UMN, LMN, motor units) Lab: Motor assessment - Muscle tone, muscle bulk, muscle strength; Modified Ashworth, Tardieu assessments of hypertonicity/spasticity
8	Peripheral nervous system – structures and function Lab: Dermatomes and myotomes vs. PN distributions – differential diagnosis of motor and/or sensory loss
9	Control circuits – modulation of movement – cerebellum Lab: Motor assessment – appendicular coordination assessment; balance assessment
10	Control circuits – modulation of movement – basal ganglia Lab: Assessment of motor control (cont.)
11	Vision and vestibular systems Lab: Vision assessment – CN, pathway, central processing of vision; Basic vestibular assessment

12	Autonomic nervous system Lab: testing the ANS – ex. Postural Hypotension
13	Thanksgiving holiday – Thursday Tuesday: Support systems – blood supply, CSF, Ventricular system Lab: CVA Case Studies
14	Support systems (cont.) – BBB; Thanksgiving break Thursday Lab: Neuro examination – putting it all together, Cases, OSCE Prep
15	Cerebrum – higher level functions; Neuroplasticity; neurotransmitters and receptors Lab: Cases, OSCE Prep
16	Final written exam and OSCE (Schedule TBA)